PSYC4411

AGENTS, ETHICS & LOOKING BACK

FROM USING AI TO DIRECTING IT — AND THE RESPONSIBILITIES THAT COME WITH IT

Week 13 · Final Session · Semester 1, 2026

TODAY’S PLAN

Two hours, two halves.

1. **A quick look back** — the thirteen-week arc and the skills you actually built
2. **AI agents** — how they work, and how they can help your research
3. **The ethics of modern AI** — in research, in education, and in society
4. **Open discussion** — the second hour is yours: what worked, what didn’t, plus the post-class survey

> The first hour is a lecture. The **second hour is a conversation** — we genuinely want your honest take on this course.

WHERE WE’VE BEEN

Thirteen weeks in one breath

THE JOURNEY

Foundations
W1 · the ML mindset; prediction vs explanation

→

Regression
W2–4 · exploring data, predicting a number

→

Classification
W5–6 · logistic regression, trees, forests

→

Structure
W7–8 · clustering & dimensionality reduction

→

Neural nets
W9–10 · how networks learn; PyTorch

→

Embeddings & LLMs
W11 · meaning in vectors

→

Viva
W12 · defending what you did

Through every week, one constant: **AI as a collaborator you direct and check** — not an oracle you trust blindly.

THE SKILLS THAT OUTLAST THE TOOLS

The LLM Problem-Solving Loop

Outer: Plan → Execute → Evaluate → Document

Inner: Engineer → Plan → Generate → Verify → Refine

Two core crafts

Prompt engineering — asking clearly enough to get useful output

Context engineering — giving the model the right information to work with

The specific models will change — probably within months. The **loop is what lasts**: plan, generate, and above all **verify**.

THE AI SKILLS YOU BUILT

Most weeks added a new way of working with AI — a ladder you've been climbing all semester:

Week	Skill	Core idea
2	Prompting	Give the AI enough context to write good code
4	Debugging	Share the full error context, not just the message
6	Refactoring	Make working code cleaner and easier to read
8	Documentation	Write clear methods descriptions for your analysis
10	Complex debugging	Diagnose silent failures from behaviour, not errors
13	**Directing agents**	**Set the goal, supervise the loop, and verify the output**

Today's capstone skill: **directing and verifying autonomous agents** — the natural next step once the AI can act on its own.

FROM *USING* AI TO *DIRECTING* IT

So far — single steps

You write a prompt

It returns code or an answer

You run it and check the result

You decide the next step yourself

Today — agents

You give a **goal**

The system plans the steps

It acts, checks, and **repeats on its own**

You **supervise** rather than drive each step

AI AGENTS: FROM ANSWERING TO ACTING

What happens when AI can take steps on its own

WHAT IS AN AGENT?

A chatbot

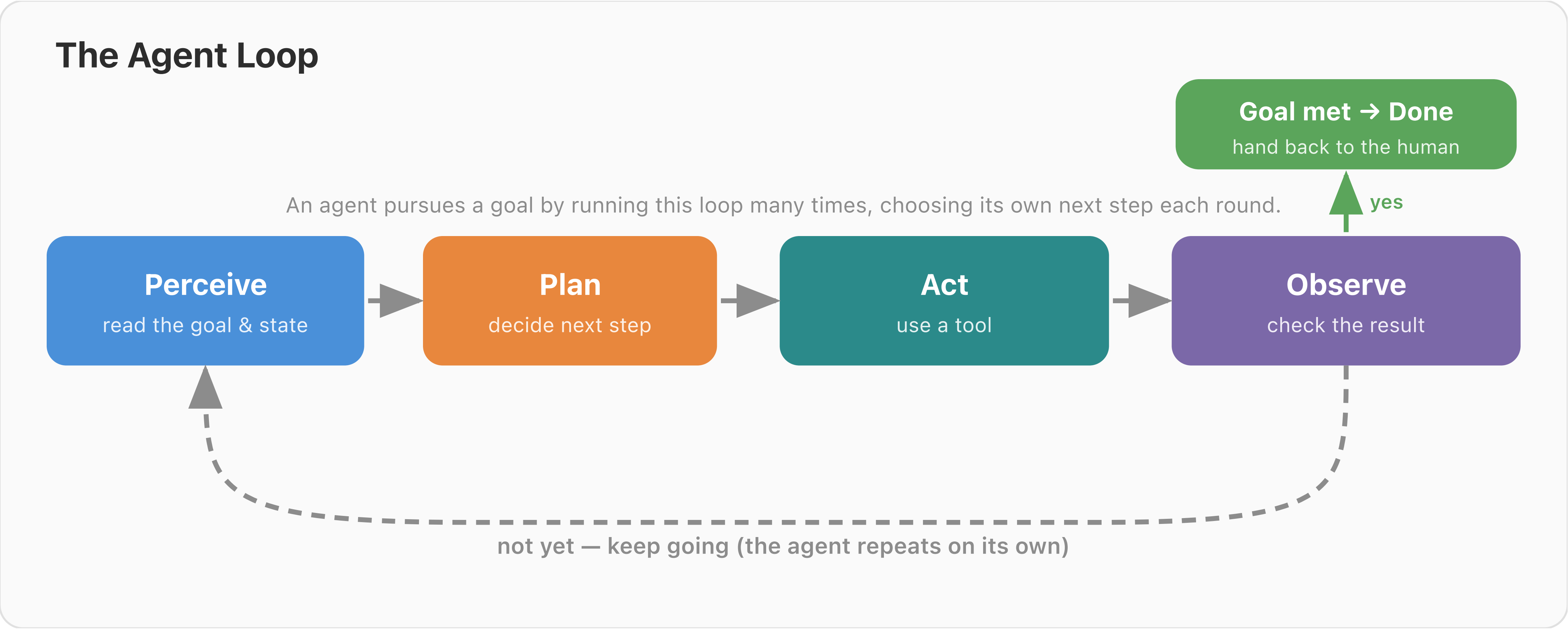
You ask → it answers → done. **One turn.**

An agent

You give a goal → it plans, acts, and checks → over **many turns**, until the goal is met.

An agent is just **a language model + tools + a loop + memory**, pointed at a goal. Nothing magic — but the loop changes what's possible.

THE AGENT LOOP



The same loop you have been running by hand — the agent now runs it for itself, deciding each next step.

TOOLS: HOW AN AGENT *ACTS*

On its own, a language model can only produce text. **Tools** let it actually do things:

- **Search** the web or a literature database
- **Run code** — clean data, fit a model, make a plot
- **Read and write files** — datasets, notes, a draft report
- **Call other software** through APIs (e.g. a survey platform, a stats package)

> Without tools, an agent can only *talk*. With tools, it can **investigate, compute, and produce** — and then read back what happened.

MEMORY & CONTEXT

Short-term

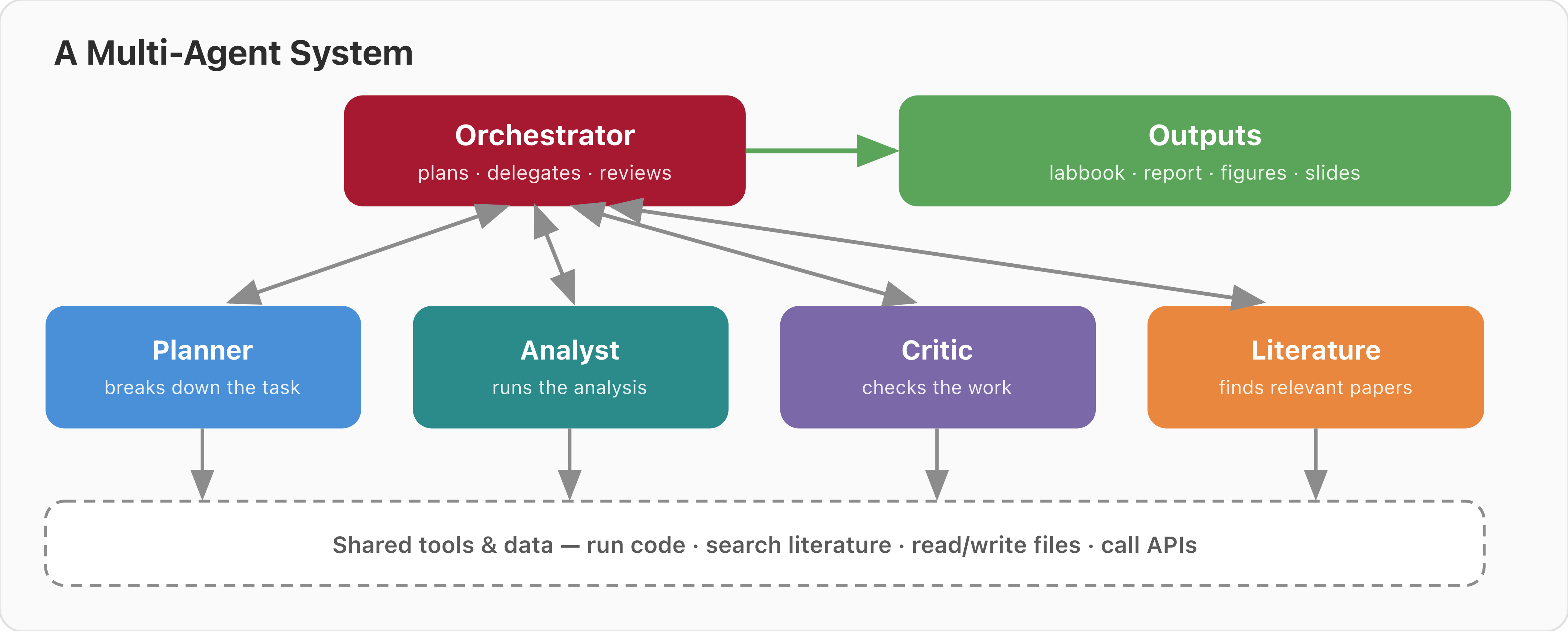
The **context window** — what the model can "see" right now. Limited, like working memory.

Long-term

Notes, files, or a database the agent can **write to and re-read later** — so it doesn't forget across many steps.

This is **context engineering** again (Weeks 1 & 11) — now the agent manages its own context as it works.

MANY AGENTS, ONE JOB



Split a big task across specialists — and let a **critic** agent check the work. It mirrors how a research team divides labour.

AGENTS FOR BEHAVIOURAL RESEARCH

Literature review
Search, screen, and synthesise papers into a cited summary you then check.

Data & analysis pipelines
Clean a messy dataset, fit several models, and compare them — end to end.

Model sweeps
Automate the “beat the baseline” search across many model variants.

Drafting outputs
Generate figures, tables, and first-draft methods sections for you to revise.

Some teams even use agents to **simulate participants** for piloting a design — useful, but treat it with real caution (it is not data about real people).

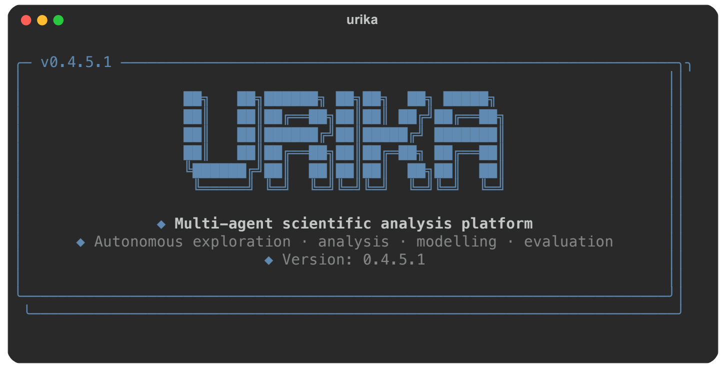
A MINI EXAMPLE: "ANALYSE THIS AND WRITE IT UP"

You hand an agent a dataset and a research question. It might:

1. Plan the analysis
2. Search the literature for relevant methods and findings
3. Clean the data and check assumptions
4. Fit several models and compare them
5. Evaluate honestly — on held-out data
6. Draft a report with figures and a methods section

> ...with a **human checkpoint** before anything is trusted. The agent does the legwork; **you remain the scientist**.

URIKA: AGENTS FOR SCIENCE (OPEN SOURCE)

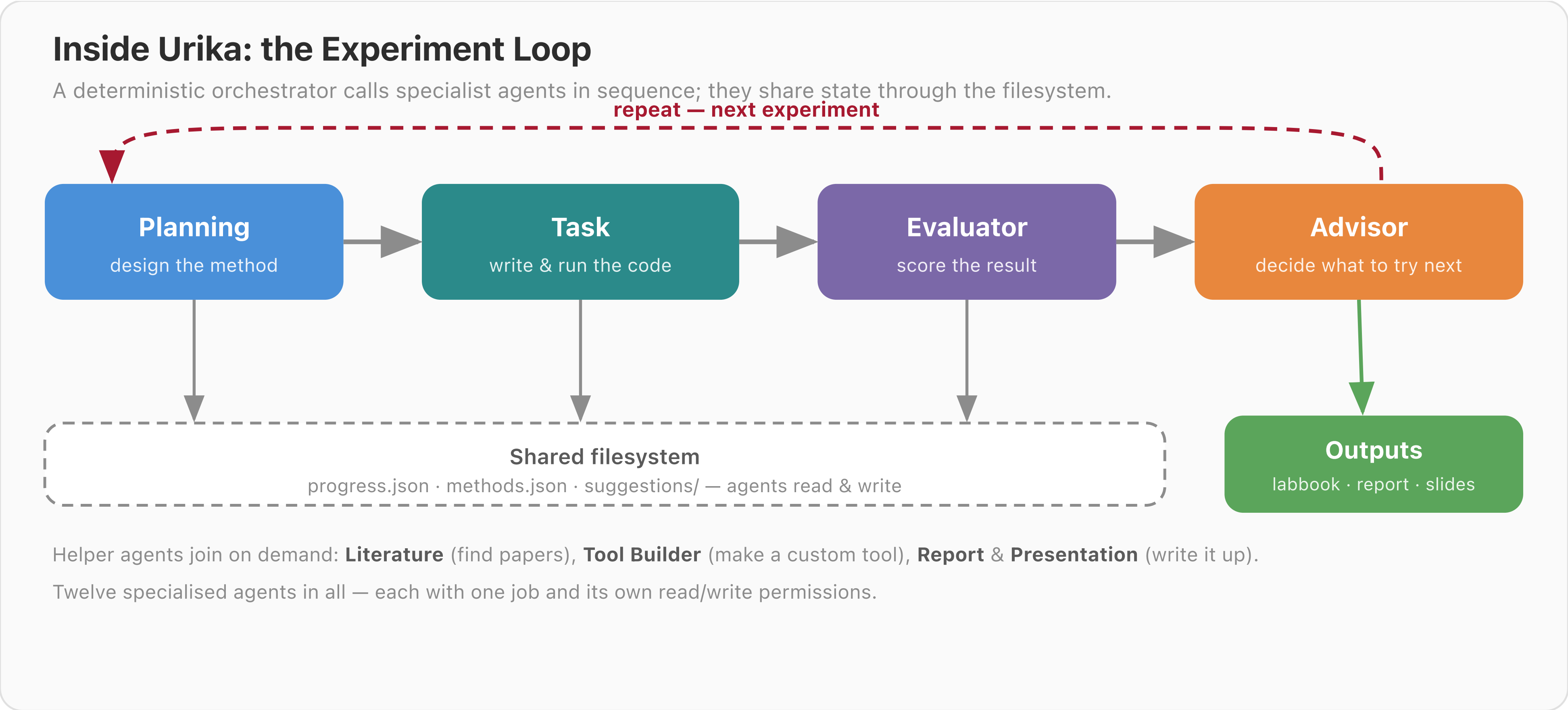


Urika is a multi-agent scientific analysis platform — from our own lab (`xkiwilabs/Urika`).

- Creates experiments, tries different methods, and **evaluates the results**
- Searches the relevant literature and **builds custom tools** when it needs them
- Documents everything automatically — **labbooks, reports, key findings, and slide decks**

> It is exactly the **loop** and the **multi-agent** ideas from the last few slides, made real for research.

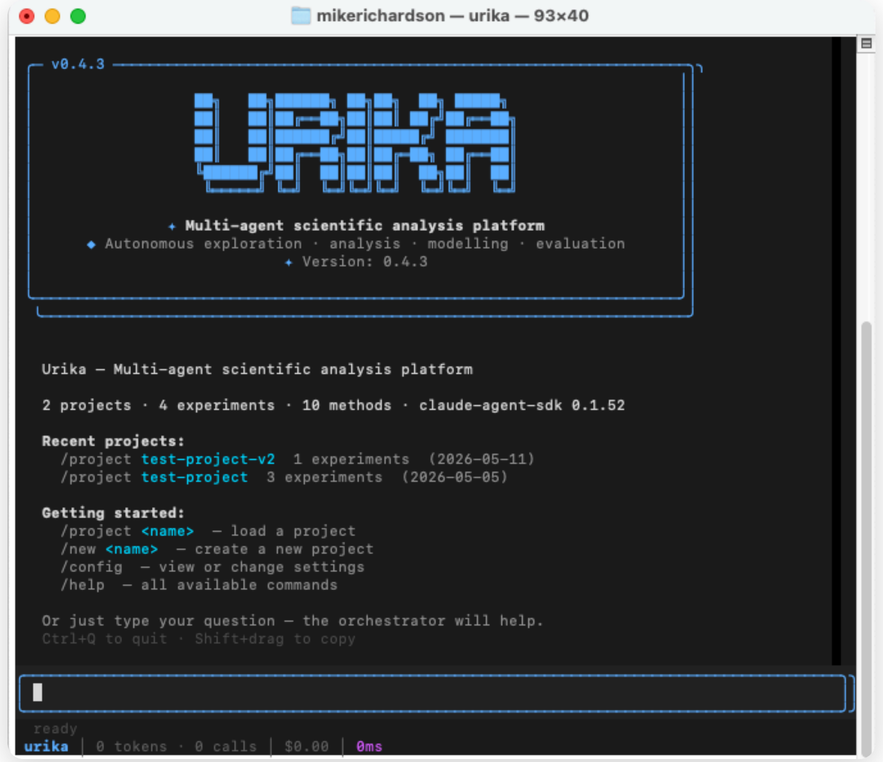
INSIDE URIKA



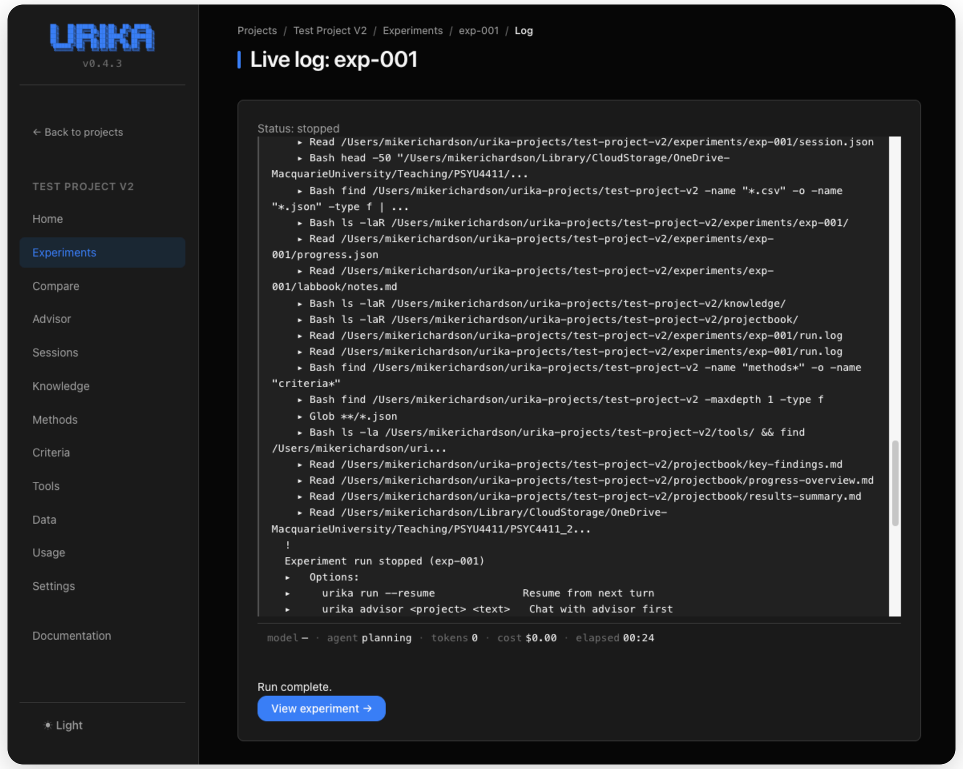
The same loop you learned — planning, acting, evaluating, deciding what's next — run by a team of specialist agents.

URIKA, IN PRACTICE

Three interfaces over the same project — a **CLI** for scripts and batch jobs, plus:



TUI — chat with the orchestrator & watch a run live



Dashboard — monitor long runs & share results in a browser

Cross-platform · runs on **local, private models** via Ollama · early development · `github.com/xkiwilabs/Urika`

OTHER OPEN-SOURCE OPTIONS

GPT-Researcher
Autonomous web & literature research that returns a cited report.

OpenHands
An autonomous coding agent — writes and runs code to complete a task.

smolagents
A lightweight Hugging Face framework for building your own simple agents.

Urika
Multi-agent science — the research-focused option we just saw.

Most are free and open — you can try them today. Start small, and always keep a human in the loop.

A REALITY CHECK

Agents are powerful. They are also **fallible in ways that matter**:

- They **hallucinate** — including confident, fake citations
- Small errors **compound** across many autonomous steps
- They cost real **money, time, and compute**
- They can act on a misunderstanding faster than you can catch it

> The more autonomy you give, the **more important the Verify step becomes** — not less. A human stays accountable for the result.

THINK ABOUT IT

Picture a research task of **your own**.

Which part would you happily hand to an agent — and which part would you **never** hand over, no matter how good it got?

THE HARDER QUESTIONS

The ethics of modern AI

THREE ARENAS

Research
Integrity, authorship, privacy, reproducibility.

Education
Learning vs offloading; fairness of access.

Society
Bias, labour, environment, power, truth.

The through-line for all three: **capability is not the same as permission**. Just because AI *can* doesn't mean you *should*.

RESEARCH INTEGRITY (1): HONESTY

- **Authorship & accountability** — AI is a tool, not an author. *You* sign off on the work and are responsible for it.
- **Transparency** — disclose how you used AI. This course models that: every commit carries a `Co-Authored-By` line.
- **Hallucinated citations** — AI invents plausible-looking references. Every citation must be checked against the real source.

> A fabricated reference in your thesis is still **your** fabricated reference — “the AI wrote it” is not a defence.

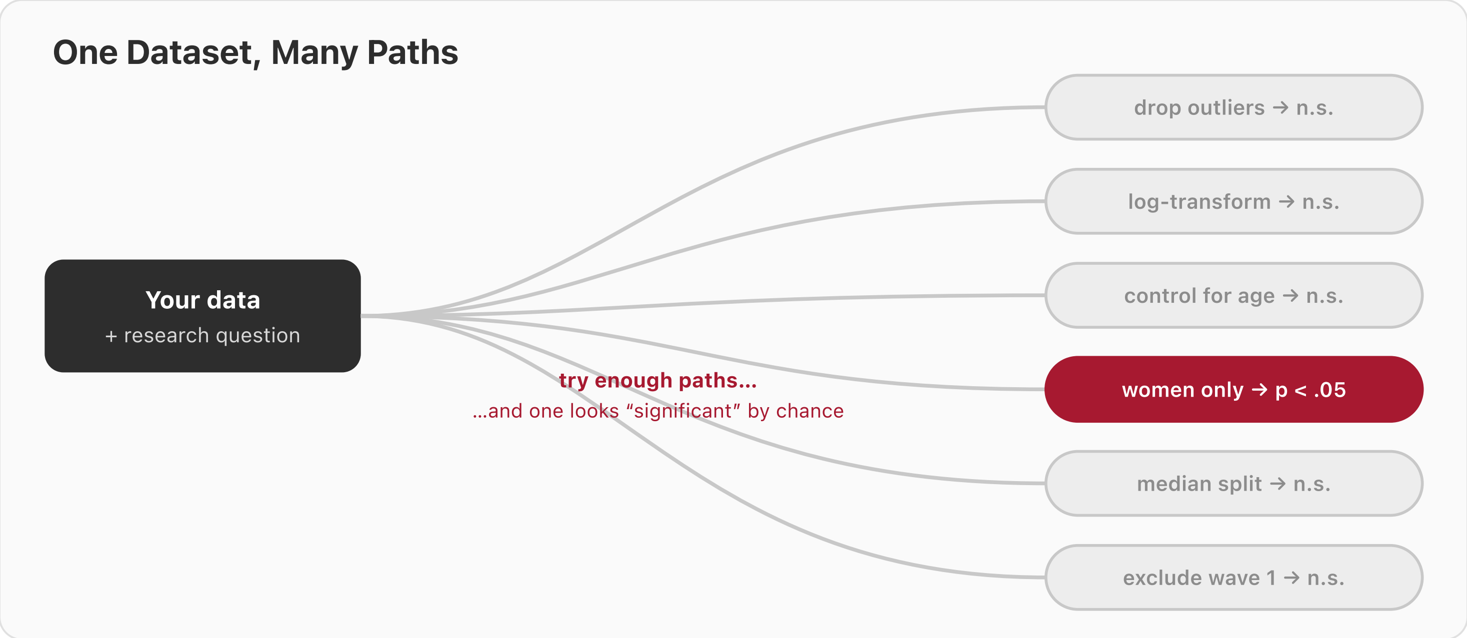
RESEARCH INTEGRITY (2): DATA & REPRODUCIBILITY

- **Data privacy** — never paste participant data into a cloud model without consent and ethics approval. For sensitive data, use **local/private models**.
- **Reproducibility** — prompts, model versions, and random seeds all drift over time. Record them, or your “method” can’t be repeated.

> “I asked ChatGPT” is not a method. **Which model, which version, what prompt, what data** — that is a method.

AGENTIC P-HACKING

An agent can run **thousands of analyses** while you sleep — the garden of forking paths (Week 3) on fast-forward.



So automation makes **pre-registration and held-out test sets matter *more*, not less**. Decide your analysis *before* you let the agent loose.

EDUCATION: LEARNING VS OFFLOADING

- **Productive struggle** — if AI does all the thinking, you don't build the understanding. The hard part is often where the learning lives.
- **Academic integrity** — using AI to *learn* differs from using it to *fake* learning.
- **Equity of access** — the best tools cost money. Who gets left behind?
- **Augment, don't atrophy** — keep the judgment; let AI take the busywork.

> This course's bet: learn **with** AI from day one, while keeping the judgment to know when it's wrong.

SOCIETY: THE WIDER PICTURE

Bias & fairness
Models learn our biases from data (Week 5) — and can scale them up.

Labour
Which jobs change, which disappear, and who decides?

Environment
Training and running large models uses real energy and water.

Truth & power
Misinformation, deepfakes, and a few firms holding the capability.

As researchers, you will help shape how these tools are used — and studied.

RESPONSIBLE AI FOR RESEARCH: A CHECKLIST

- ✅ **Disclose** how you used AI
- ✅ **Verify** every claim and every citation
- ✅ **Protect** participant data — consent, ethics, private models
- ✅ Keep a **human accountable** for the result
- ✅ **Pre-register** the analysis you intend to run
- ✅ **Record** the model, version, and prompts

> None of this is new ethics — it's **good science**, applied honestly to a powerful new tool.

THINK ABOUT IT

Where is **your** line?

What use of AI in research feels clearly fine to you — and what feels clearly over the line? What sits uncomfortably in between?

Hold that thought — it's where our discussion begins.

YOUR TURN

An open conversation with the teaching team & guests

HOW DID THIS GO?

No right answers — we want the honest version.

- What **worked** for you this semester — and what **didn’t**?
- What **surprised** you about learning to use AI for research?
- The **LLM-assisted format** — did it help your learning, or get in the way?
- Has your **confidence** with data and AI shifted? In which direction?
- What would you **change** for next year’s cohort?

POST-CLASS SURVEY

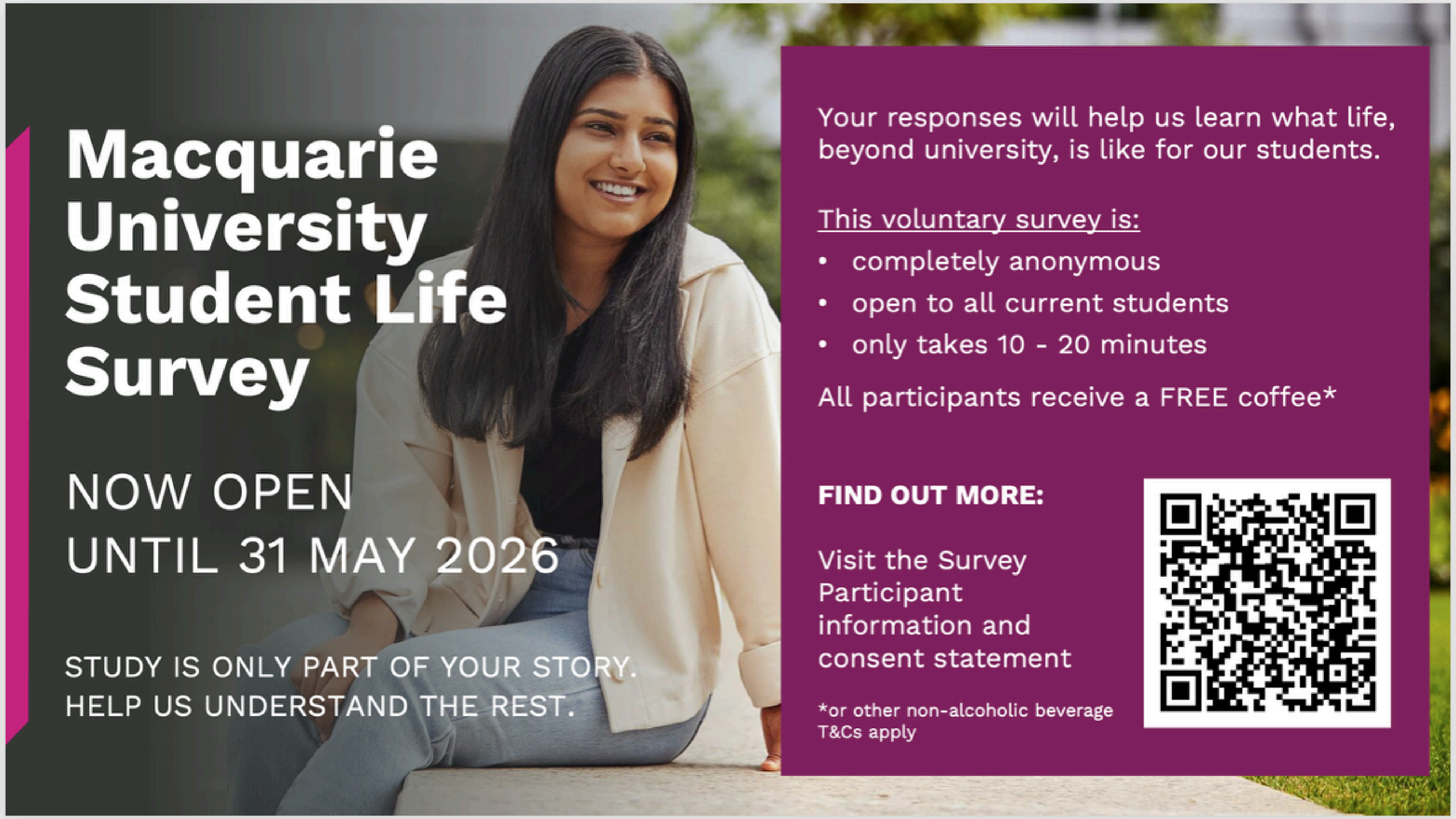
Your feedback directly shapes next year's course. Please take a few minutes now.

Open the survey

Anonymous · takes about 5 minutes · `mqedu.qualtrics.com/jfe/form/SV_83047akpZ0xSTYy`

AND ONE MORE: THE MQ STUDENT LIFE SURVEY

While you're at it — Macquarie's university-wide student survey. Scan the QR code on the flyer to take part.



THANK YOU

You came in having (mostly) never written a line of code. You leave as **AI-literate researchers**.

Keep running the loop: Plan → Prompt → Verify → Refine → Document.

Course repo · `xkiwilabs/Urika` · this week's readings